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║         THE COMPLETE DSA PATTERN SHEET                       ║
║         Every Pattern You Need to Know                       ║
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> *Study one pattern at a time. Consistency beats intensity.*

Legend

Symbol	Difficulty
`[E]`	Easy
`[M]`	Medium
`[H]`	Hard

Table of Contents

#	Topic
12	Greedy
13	Dynamic Programming
14	Graph
15	Tries

TOPIC 1 — BASICS

｡★ Pattern: Conditionals (if-else)

#	Problem	Difficulty	Done?
1	Fizz Buzz	[E]	- [ ]
2	Palindrome Number	[E]	- [ ]
3	Number of Steps to Reduce a Number to Zero	[E]	- [ ]

｡★ Pattern: Loops

#	Problem	Difficulty	Done?
1	Running Sum of 1d Array	[E]	- [ ]
2	Find Numbers with Even Number of Digits	[E]	- [ ]
3	Richest Customer Wealth	[E]	- [ ]

｡★ Pattern: Simulation / Implementation

#	Problem	Difficulty	Done?
1	Baseball Game	[E]	- [ ]
2	Robot Return to Origin	[E]	- [ ]

｡★ Pattern: Maths

#	Problem	Difficulty	Done?
1	Reverse Integer	[M]	- [ ]
2	Plus One	[E]	- [ ]
3	Add Digits	[E]	- [ ]

TOPIC 2 — ARRAYS

｡★ Pattern: Fundamentals

#	Problem	Difficulty	Done?
1	Rotate Array	[M]	- [x]
2	Move Zeroes	[E]	- [ ]
3	Remove Duplicates from Sorted Array	[E]	- [ ]

｡★ Pattern: Prefix Sums

#	Problem	Difficulty	Done?
1	Continuous Subarray Sum	[M]	- [ ]
2	Subarray Sum Equals K	[M]	- [ ]
3	Product of Array Except Self	[M]	- [ ]
4	Subarray Sums Divisible by K	[M]	- [ ]
5	Range Sum Query 2D - Immutable	[M]	- [ ]

｡★ Pattern: Kadane's Algorithm

#	Problem	Difficulty	Done?
1	Maximum Subarray	[M]	- [ ]
2	Maximum Product Subarray	[M]	- [ ]
3	Best Time to Buy and Sell Stock	[M]	- [ ]

◦ ★ Pattern: Intervals

#	Problem	Difficulty	Done?
1	Merge Intervals	[M]	- [ ]
2	Insert Interval	[M]	- [ ]
3	Non-overlapping Intervals	[M]	- [ ]

◦ ★ Pattern: Hashing on Arrays

#	Problem	Difficulty	Done?
1	Two Sum	[E]	- [ ]
2	Majority Element	[E]	- [ ]
3	Majority Element II	[M]	- [ ]
4	Top K Frequent Elements	[M]	- [ ]
5	Valid Anagram	[E]	- [ ]
6	Group Anagrams	[M]	- [ ]
7	Longest Consecutive Sequence	[M]	- [ ]

◦ ★ Pattern: 2D Arrays / Matrix

#	Problem	Difficulty	Done?
1	Spiral Matrix	[M]	- [ ]
2	Spiral Matrix II	[M]	- [ ]
3	Rotate Image	[M]	- [ ]

#	Problem	Difficulty	Done?
4	Set Matrix Zeroes	[M]	- [ ]

TOPIC 3 — STRINGS

｡★ Pattern: Fundamentals

#	Problem	Difficulty	Done?
1	Reverse String	[E]	- [ ]
2	Reverse Words in a String III	[E]	- [ ]
3	Roman to Integer	[E]	- [ ]
4	String Compression	[M]	- [ ]

｡★ Pattern: Frequency & Hashing

#	Problem	Difficulty	Done?
1	Isomorphic Strings	[E]	- [ ]
2	Bulls and Cows	[M]	- [ ]

｡★ Pattern: Palindromes

#	Problem	Difficulty	Done?
1	Valid Palindrome	[E]	- [ ]
2	Longest Palindromic Substring	[M]	- [ ]
3	Palindromic Substrings	[M]	- [ ]
4	Palindrome Partitioning	[M]	- [ ]

｡★ Pattern: Simulation

#	Problem	Difficulty	Done?
1	Zigzag Conversion	[M]	- [ ]
2	Multiply Strings	[M]	- [ ]
3	Decode String	[M]	- [ ]

⭐ Pattern: Prefix/Suffix & Pattern Matching

#	Problem	Difficulty	Done?
1	Longest Common Prefix	[E]	- [ ]
2	Find and Replace Pattern	[M]	- [ ]
3	Remove Duplicate Letters	[M]	- [ ]

TOPIC 4 — RECURSION & BACKTRACKING

⭐ Pattern: Fundamentals

#	Problem	Difficulty	Done?
1	Fibonacci Number	[E]	- [ ]
2	Power of Two	[E]	- [ ]
3	Pow(x, n)	[M]	- [ ]

⭐ Pattern: Recursion

#	Problem	Difficulty	Done?
1	Different Ways to Add Parentheses	[M]	- [ ]
2	K-th Symbol in Grammar	[M]	- [ ]
3	Beautiful Arrangement	[M]	- [ ]

#	Problem	Difficulty	Done?
4	Parsing A Boolean Expression	[H]	- [ ]
5	Integer to English Words	[H]	- [ ]

◦★ Pattern: Backtracking

#	Problem	Difficulty	Done?
1	Subsets	[M]	- [ ]
2	Subsets II	[M]	- [ ]
3	Permutations	[M]	- [ ]
4	Permutations II	[M]	- [ ]
5	Combination Sum	[M]	- [ ]
6	Combination Sum II	[M]	- [ ]
7	Combination Sum III	[M]	- [ ]
8	N-Queens	[H]	- [ ]
9	N-Queens II	[H]	- [ ]
10	Sudoku Solver	[H]	- [ ]

TOPIC 5 — TWO POINTERS

◦★ Pattern: Opposite Ends (Left <-> Right)

#	Problem	Difficulty	Done?
1	Two Sum II - Input Array Is Sorted	[M]	- [ ]
2	Container With Most Water	[M]	- [ ]
3	Trapping Rain Water	[H]	- [ ]

｡★ Pattern: Merge Two Sorted Array / Sequence

#	Problem	Difficulty	Done?
1	Merge Sorted Array	[E]	- [ ]
2	Interval List Intersections	[M]	- [ ]

｡★ Pattern: Fixed + Two Pointers

#	Problem	Difficulty	Done?
1	3Sum	[M]	- [ ]
2	3Sum Closest	[M]	- [ ]
3	4Sum	[M]	- [ ]

TOPIC 6 — SLIDING WINDOW

｡★ Pattern: Fixed Size Window

#	Problem	Difficulty	Done?
1	Maximum Average Subarray I	[E]	- [ ]
2	Number of Sub-Arrays of Size K and Average >= Threshold	[M]	- [ ]
3	Maximum Sum of Distinct Subarrays With Length K	[M]	- [ ]

｡★ Pattern: Variable Size Window

#	Problem	Difficulty	Done?
1	Longest Subarray of 1's After Deleting One Element	[M]	- [ ]
2	Max Consecutive Ones III	[M]	- [ ]

#	Problem	Difficulty	Done?
3	Fruit Into Baskets	[M]	- [ ]
4	Binary Subarrays With Sum	[M]	- [ ]
5	Count Number of Nice Subarrays	[M]	- [ ]
6	Subarray Product Less Than K	[M]	- [ ]
7	Subarrays With K Different Integers	[H]	- [ ]

｡★ Pattern: Sliding Window on Strings

#	Problem	Difficulty	Done?
1	Maximum Number of Vowels in a Substring of Given Length	[M]	- [ ]
2	Longest Substring Without Repeating Characters	[M]	- [ ]
3	Longest Repeating Character Replacement	[M]	- [ ]
4	Find All Anagrams in a String	[M]	- [ ]
5	Minimum Window Substring	[H]	- [ ]

TOPIC 7 — STACK & QUEUES

｡★ Pattern: Implementation

#	Problem	Difficulty	Done?
1	Min Stack	[M]	- [ ]
2	Design a Stack With Increment Operation	[M]	- [ ]
3	Design Circular Queue	[M]	- [ ]
4	Design Front Middle Back Queue	[M]	- [ ]

｡★ Pattern: Expression Evaluation

#	Problem	Difficulty	Done?
1	Evaluate Reverse Polish Notation	[M]	- [ ]
2	Number of Atoms	[H]	- [ ]
3	Basic Calculator	[H]	- [ ]

｡⋆ Pattern: Parentheses Processing

#	Problem	Difficulty	Done?
1	Longest Valid Parentheses	[H]	- [ ]
2	Minimum Remove to Make Valid Parentheses	[M]	- [ ]
3	Minimum Add to Make Parentheses Valid	[M]	- [ ]

｡⋆ Pattern: Monotonic Stacks

#	Problem	Difficulty	Done?
1	Next Greater Element I	[E]	- [ ]
2	Next Greater Element II	[M]	- [ ]
3	Largest Rectangle in Histogram	[H]	- [ ]
4	Maximal Rectangle	[H]	- [ ]
5	Daily Temperatures	[M]	- [ ]
6	Sum of Subarray Minimums	[M]	- [ ]
7	Sum of Subarray Ranges	[M]	- [ ]
8	Asteroid Collision	[M]	- [ ]

TOPIC 8 — LINKED LIST

｡⋆ Pattern: Fast & Slow Pointers

#	Problem	Difficulty	Done?
1	Middle of the Linked List	[E]	- [ ]
2	Linked List Cycle	[E]	- [ ]
3	Linked List Cycle II	[M]	- [ ]

｡⋆ Pattern: Node Rearrangements

#	Problem	Difficulty	Done?
1	Remove Nth Node from End of List	[M]	- [ ]
2	Swap Nodes in Pairs	[M]	- [ ]
3	Odd Even Linked List	[M]	- [ ]

｡⋆ Pattern: Reversal

#	Problem	Difficulty	Done?
1	Reverse Linked List	[E]	- [ ]
2	Reverse Linked List II	[M]	- [ ]
3	Reverse Nodes in k-Group	[M]	- [ ]

｡⋆ Pattern: Merge & Multiple Lists

#	Problem	Difficulty	Done?
1	Merge Two Sorted Lists	[E]	- [ ]
2	Merge k Sorted Lists	[H]	- [ ]
3	Intersection of Two Linked Lists	[E]	- [ ]
4	Add Two Numbers	[M]	- [ ]

TOPIC 9 — TREES

◦★ Pattern: Traversal (BFS / DFS)

#	Problem	Difficulty	Done?
1	Binary Tree Level Order Traversal	[M]	- [ ]
2	Binary Tree Zigzag Level Order Traversal	[M]	- [ ]
3	Add One Row to Tree	[M]	- [ ]

◦★ Pattern: Depth / Height Based

#	Problem	Difficulty	Done?
1	Maximum Depth of Binary Tree	[E]	- [ ]
2	Minimum Depth of Binary Tree	[E]	- [ ]
3	Maximum Depth of N-ary Tree	[E]	- [ ]

◦★ Pattern: Comparison

#	Problem	Difficulty	Done?
1	Same Tree	[E]	- [ ]
2	Symmetric Tree	[E]	- [ ]
3	Leaf-Similar Trees	[E]	- [ ]

◦★ Pattern: Root to Leaves

#	Problem	Difficulty	Done?
1	Sum Root to Leaf Numbers	[M]	- [ ]
2	Sum of Root to Leaf Binary Numbers	[E]	- [ ]
3	Binary Tree Paths	[E]	- [ ]
4	Path Sum	[E]	- [ ]

｡⋆ Pattern: Ancestor

#	Problem	Difficulty	Done?
1	Lowest Common Ancestor of a Binary Tree	[M]	- [ ]
2	Lowest Common Ancestor of Deepest Leaves	[M]	- [ ]
3	Kth Ancestor of a Tree Node	[H]	- [ ]
4	Maximum Difference Between Node and Ancestor	[M]	- [ ]

｡⋆ Pattern: Binary Search Tree (BST)

#	Problem	Difficulty	Done?
1	Search in a Binary Search Tree	[E]	- [ ]
2	Validate Binary Search Tree	[M]	- [ ]
3	Convert Sorted Array to Binary Search Tree	[E]	- [ ]
4	Minimum Absolute Difference in BST	[E]	- [ ]
5	Kth Smallest Element in a BST	[M]	- [ ]
6	Lowest Common Ancestor of a Binary Search Tree	[M]	- [ ]

TOPIC 10 — BINARY SEARCH

｡⋆ Pattern: Classic Binary Search on Sorted Arrays

#	Problem	Difficulty	Done?
1	Binary Search	[E]	- [ ]
2	Search Insert Position	[E]	- [ ]
3	Find First and Last Position of Element in Sorted Array	[M]	- [ ]
4	Median of Two Sorted Arrays	[H]	- [ ]

｡⋆ Pattern: Binary Search on Answer

#	Problem	Difficulty	Done?
1	Koko Eating Bananas	`[M]`	- [ ]
2	Capacity to Ship Packages Within D Days	`[M]`	- [ ]
3	Minimum Number of Days to Make m Bouquets	`[M]`	- [ ]
4	Maximum Running Time of N Computers	`[H]`	- [ ]

｡⋆ Pattern: Binary Search on Rotated / Modified Sorted Arrays

#	Problem	Difficulty	Done?
1	Search in Rotated Sorted Array	`[M]`	- [ ]
2	Search in Rotated Sorted Array II	`[M]`	- [ ]
3	Find Minimum in Rotated Sorted Array	`[M]`	- [ ]
4	Find Minimum in Rotated Sorted Array II	`[H]`	- [ ]

｡⋆ Pattern: Binary Search on Matrix

#	Problem	Difficulty	Done?
1	Search a 2D Matrix	`[M]`	- [ ]
2	Search a 2D Matrix II	`[M]`	- [ ]
3	Kth Smallest Element in a Sorted Matrix	`[M]`	- [ ]

TOPIC 11 — HEAP (PRIORITY QUEUE)

｡⋆ Pattern: Top K

#	Problem	Difficulty	Done?
1	Kth Largest Element in an Array	[M]	- [ ]
2	Top K Frequent Words	[M]	- [ ]
3	K Closest Points to Origin	[M]	- [ ]

◦ ★ Pattern: Merge K Sorted

#	Problem	Difficulty	Done?
1	Find K Pairs With Smallest Sums	[M]	- [ ]
2	Smallest Range Covering Elements From K Lists	[H]	- [ ]

◦ ★ Pattern: Two Heaps

#	Problem	Difficulty	Done?
1	Find Median From Data Stream	[H]	- [ ]
2	IPO	[H]	- [ ]
3	Sliding Window Median	[H]	- [ ]

◦ ★ Pattern: Finding Minimums

#	Problem	Difficulty	Done?
1	Minimum Cost to Hire K Workers	[H]	- [ ]
2	Minimum Number of Refueling Stops	[H]	- [ ]
3	Task Scheduler	[M]	- [ ]

TOPIC 12 — GREEDY

◦ ★ Pattern: Local Optimal Selection

#	Problem	Difficulty	Done?
1	Jump Game	[M]	- [ ]
2	Jump Game II	[M]	- [ ]
3	Gas Station	[M]	- [ ]
4	Wiggle Subsequence	[M]	- [ ]

｡★ Pattern: Distribution / Resource Allocation

#	Problem	Difficulty	Done?
1	Minimum Cost to Move Chips to the Same Position	[E]	- [ ]
2	Assign Cookies	[E]	- [ ]
3	Candy	[H]	- [ ]

｡★ Pattern: Number Construction

#	Problem	Difficulty	Done?
1	Remove K Digits	[M]	- [ ]
2	Monotone Increasing Digits	[M]	- [ ]

｡★ Pattern: Sort then Apply Greedy

#	Problem	Difficulty	Done?
1	Queue Reconstruction by Height	[M]	- [ ]
2	Boats to Save People	[M]	- [ ]
3	Two City Scheduling	[M]	- [ ]
4	Car Pooling	[M]	- [ ]

TOPIC 13 — DYNAMIC PROGRAMMING

｡★ Pattern: 1D DP

#	Problem	Difficulty	Done?
1	Climbing Stairs	[E]	- [ ]
2	House Robber	[M]	- [ ]
3	House Robber II	[M]	- [ ]
4	Coin Change	[M]	- [ ]
5	Coin Change II	[M]	- [ ]
6	Decode Ways	[M]	- [ ]
7	Combination Sum IV	[M]	- [ ]

｡★ Pattern: Grid DP

#	Problem	Difficulty	Done?
1	Unique Paths	[M]	- [ ]
2	Unique Paths II	[M]	- [ ]
3	Minimum Path Sum	[M]	- [ ]
4	Maximal Square	[M]	- [ ]
5	Count Square Submatrices With All Ones	[M]	- [ ]
6	Dungeon Game	[M]	- [ ]

｡★ Pattern: Knapsack

#	Problem	Difficulty	Done?
1	Target Sum	[M]	- [ ]
2	Ones and Zeroes	[M]	- [ ]
3	Partition Equal Subset Sum	[M]	- [ ]
4	Partition Array Into Two Arrays To Minimize Sum Difference	[H]	- [ ]

#	Problem	Difficulty	Done?
5	Find the Sum of the Power of All Subsequences	[H]	- [ ]

◦★ Pattern: LCS (Longest Common Subsequence)

#	Problem	Difficulty	Done?
1	Longest Common Subsequence	[M]	- [ ]
2	Longest Palindromic Subsequence	[M]	- [ ]
3	Edit Distance	[M]	- [ ]
4	Shortest Common Supersequence	[H]	- [ ]
5	Regular Expression Matching	[H]	- [ ]
6	Distinct Subsequences	[H]	- [ ]
7	Wildcard Matching	[H]	- [ ]

◦★ Pattern: LIS (Longest Increasing Subsequence)

#	Problem	Difficulty	Done?
1	Longest Increasing Subsequence	[M]	- [ ]
2	Number of Longest Increasing Subsequence	[M]	- [ ]
3	Longest Arithmetic Subsequence	[M]	- [ ]
4	Largest Divisible Subset	[M]	- [ ]
5	Make Array Strictly Increasing	[H]	- [ ]
6	Russian Doll Envelopes	[H]	- [ ]

◦★ Pattern: Palindrome DP

#	Problem	Difficulty	Done?
1	Palindrome Partitioning	[M]	- [ ]
2	Palindrome Partitioning II	[H]	- [ ]
3	Minimum Insertion Steps to Make a String Palindrome	[H]	- [ ]
4	Count Different Palindromic Subsequences	[H]	- [ ]

｡★ Pattern: Stock DP

#	Problem	Difficulty	Done?
1	Best Time to Buy and Sell Stock	[E]	- [ ]
2	Best Time to Buy and Sell Stock II	[M]	- [ ]
3	Best Time to Buy and Sell Stock III	[H]	- [ ]
4	Best Time to Buy and Sell Stock IV	[H]	- [ ]
5	Best Time to Buy and Sell Stock With Cooldown	[M]	- [ ]
6	Best Time to Buy and Sell Stock With Transaction Fee	[M]	- [ ]

｡★ Pattern: Partition DP

#	Problem	Difficulty	Done?
1	Stone Game	[M]	- [ ]
2	Stone Game II	[M]	- [ ]
3	Stone Game III	[H]	- [ ]
4	Minimum Cost to Cut a Stick	[H]	- [ ]
5	Burst Balloons	[H]	- [ ]
6	Remove Boxes	[H]	- [ ]

｡★ Pattern: Tree DP

#	Problem	Difficulty	Done?
1	House Robber III	[M]	- [ ]
2	Binary Tree Maximum Path Sum	[H]	- [ ]
3	Sum of Distances in Tree	[H]	- [ ]

｡★ Pattern: Bitmask DP

#	Problem	Difficulty	Done?
1	Can I Win	[M]	- [ ]
2	Partition to K Equal Sum Subsets	[M]	- [ ]
3	Shortest Path Visiting All Nodes	[H]	- [ ]

TOPIC 14 — GRAPH

｡★ Pattern: Graph Traversal

#	Problem	Difficulty	Done?
1	Flood Fill	[E]	- [ ]
2	Keys and Rooms	[M]	- [ ]
3	Number of Islands	[M]	- [ ]
4	Surrounded Regions	[M]	- [ ]
5	Pacific Atlantic Water Flow	[M]	- [ ]
6	Making a Large Island	[H]	- [ ]

｡★ Pattern: Connected Components

#	Problem	Difficulty	Done?
1	Number of Provinces	[M]	- [ ]

#	Problem	Difficulty	Done?
2	Number of Operations to Make Network Connected	[M]	- [ ]
3	Minimum Number of Days to Disconnect Island	[H]	- [ ]

◦ ★ Pattern: Multi Source BFS

#	Problem	Difficulty	Done?
1	01 Matrix	[M]	- [ ]
2	Rotting Oranges	[M]	- [ ]
3	As Far from Land as Possible	[M]	- [ ]
4	Escape the Spreading Fire	[H]	- [ ]

◦ ★ Pattern: Topological Sort

#	Problem	Difficulty	Done?
1	Course Schedule	[M]	- [ ]
2	Course Schedule II	[M]	- [ ]
3	Find Eventual Safe States	[M]	- [ ]

◦ ★ Pattern: Union Find

#	Problem	Difficulty	Done?
1	Redundant Connection	[M]	- [ ]
2	Accounts Merge	[M]	- [ ]
3	Checking Existence of Edge Length Limited Paths	[H]	- [ ]
4	Remove Max Number of Edges to Keep Graph Fully Traversable	[H]	- [ ]

◦ ★ Pattern: Shortest Path

#	Problem	Difficulty	Done?
1	Shortest Path in Binary Matrix	[M]	- [ ]
2	Network Delay Time	[M]	- [ ]
3	Path With Minimum Effort	[M]	- [ ]
4	Cheapest Flights Within K Stops	[M]	- [ ]
5	Number of Ways to Arrive At Destination	[M]	- [ ]

｡★ Pattern: Bipartite Graph / Coloring

#	Problem	Difficulty	Done?
1	Is Graph Bipartite?	[M]	- [ ]
2	Possible Bipartition	[M]	- [ ]

｡★ Pattern: Minimum Spanning Tree

#	Problem	Difficulty	Done?
1	Minimum Cost to Connect All Points	[M]	- [ ]

｡★ Pattern: Grid as Graph

#	Problem	Difficulty	Done?
1	Number of Closed Islands	[M]	- [ ]
2	Number of Enclaves	[M]	- [ ]
3	Swim in Rising Water	[H]	- [ ]

TOPIC 15 — TRIES

｡★ Pattern: Implementation

#	Problem	Difficulty	Done?
1	Implement Trie (Prefix Tree)	[M]	- [ ]

｡★ Pattern: Prefix / Suffix Queries

#	Problem	Difficulty	Done?
1	Prefix and Suffix Search	[H]	- [ ]
2	Longest Word in Dictionary	[M]	- [ ]
3	Longest Word in Dictionary through Deletion	[M]	- [ ]

｡★ Pattern: XOR Trie

#	Problem	Difficulty	Done?
1	Maximum XOR of Two Numbers in an Array	[M]	- [ ]
2	Maximum XOR With an Element from Array	[H]	- [ ]
3	Count Pairs With XOR in a Range	[H]	- [ ]

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